

Problem Set Lab 04 (graded), Sep. 24, 2026 (Generalization and Model Selection)

Goals. The goal of this week's lab is to:

- Focus on the theoretical content from Week 2 until now.
- Implement 4-fold cross-validation.
- Understand the bias-variance decomposition.

Submission instructions:

- Please submit a PDF file (for solutions to the graded labs) to canvas.
- Deadline: 23:59 on Oct. 8, 2026

1 Graded labs [80 pts]

Problem 1 (Huber Loss [5 pts]):

Consider a linear model $f(\mathbf{w}, \mathbf{x}_n) = \mathbf{x}_n^\top \mathbf{w}$ and the cost function

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N h_c(y_n - \mathbf{x}_n^\top \mathbf{w}), \quad h_c(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leq c, \\ c|r| - \frac{1}{2}c^2, & |r| > c, \end{cases}$$

where $c > 0$ is fixed. Compute $\nabla \mathcal{L}(\mathbf{w})$, including the cases where a residual equals c or $-c$.

Problem 2 (Maximum A Posteriori Estimation [15 pts]):

Suppose $y_n = \mathbf{x}_n^\top \mathbf{w} + \epsilon_n$, where $\mathbf{x}_n \in \mathbb{R}^D$ and the noises ϵ_n are independent $\mathcal{N}(0, \sigma^2)$ random variables. Assume a prior $\mathbf{w} \sim \mathcal{N}(\mathbf{m}, \tau^2 \mathbf{I}_D)$, where $\mathbf{m} \in \mathbb{R}^D$ is known and need not be zero. Both $\sigma^2 > 0$ and $\tau^2 > 0$ are known. Let $\mathbf{X}$ have rows $\mathbf{x}_n^\top$ and let $\mathbf{y} = (y_1, \dots, y_N)^\top$.

1. (5 pts) Write down the likelihood $p(\mathbf{y} | \mathbf{X}, \mathbf{w})$ and its negative logarithm.
2. (5 pts) Show that the MAP estimate minimizes

$$J(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \frac{\lambda}{2} \|\mathbf{w} - \mathbf{m}\|_2^2.$$

Express λ in terms of N , σ^2 , and τ^2 .

3. (5 pts) Derive the normal equations and the unique MAP estimate. Explain why the solution is unique even if $\mathbf{X}$ is rank deficient.

Problem 3 (Repeated Features and Ridge Regression [20 pts]):

For scalar inputs and outputs with $\sum_n x_n^2 > 0$, compare these two models without intercepts:

$$J_A(w) = \frac{1}{2N} \sum_{n=1}^N (y_n - wx_n)^2 + \frac{\lambda}{2} w^2,$$

$$J_B(u, v) = \frac{1}{2N} \sum_{n=1}^N (y_n - (u + v)x_n)^2 + \frac{\lambda}{2} (u^2 + v^2).$$

Model B repeats the same input feature twice. Define $s = \frac{1}{N} \sum_n x_n^2$ and $t = \frac{1}{N} \sum_n x_n y_n$.

- (5 pts) For $\lambda = 0$, find all minimizers of both objectives. Are the parameters unique? Do the two models give the same predictions?
- (5 pts) For $\lambda > 0$, derive the first-order conditions for Model B and show that its minimizer satisfies $u^* = v^*$.
- (5 pts) Find w^* for Model A and $u^* + v^*$ for Model B when $\lambda > 0$. Do their predictions necessarily coincide when they use the same λ ?
- (5 pts) Suppose Model A uses a fixed $\lambda_A > 0$. What value λ_B should Model B use to give the same predictions for every such dataset? Explain using the smallest value of $u^2 + v^2$ subject to $u + v = w$.

Problem 4 (Bias and Variance of a Shrinkage Estimator [20 pts]):

Let $y_n = \theta + \epsilon_n$, $n = 1, \dots, N$, where θ is a fixed unknown scalar and the noises are independent with mean zero and variance $\sigma^2 > 0$. Define $\bar{y} = \frac{1}{N} \sum_n y_n$ and the estimator $\hat{\theta}_a = a\bar{y}$, where $a \in [0, 1]$ is fixed independently of the training sample.

- (5 pts) Compute the bias $\mathbb{E}[\hat{\theta}_a] - \theta$ and variance $\text{Var}(\hat{\theta}_a)$.
- (5 pts) Let $y_{\text{new}} = \theta + \epsilon_{\text{new}}$, where the new noise has the same mean and variance and is independent of the training sample. Derive $R(a) = \mathbb{E}[(y_{\text{new}} - \hat{\theta}_a)^2]$, with the expectation over both the training sample and the new noise. Identify its noise, squared bias, and variance terms.
- (10 pts) Find the value $a^* \in [0, 1]$ minimizing $R(a)$ and compare $R(a^*)$ with $R(1)$. Explain why allowing bias can help, and why this expression for a^* cannot generally be used directly in practice.

Problem 5 (Generalization after Model Selection [20 pts]):

Let $f_1, \dots, f_M$ be predictors fitted using only a training set. Evaluate them on an independent validation set of n i.i.d. observations from the data distribution, using a loss in $[0, 1]$. Write L_j and $\hat{L}_j$ for their population and validation risks. Conditional on the training set, Hoeffding's inequality gives, for each fixed j ,

$$\mathbb{P}(|L_j - \hat{L}_j| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

- (5 pts) Use the union bound to find $\epsilon = \epsilon(n, M, \delta)$ such that, with probability at least $1 - \delta$, $|L_j - \hat{L}_j| \leq \epsilon$ holds simultaneously for all j . Assume $0 < \delta < 1$.

Hint: Let $A_j = \{|L_j - \hat{L}_j| \geq \epsilon\}$ and use $\mathbb{P}(\bigcup_{j=1}^M A_j) \leq \sum_{j=1}^M \mathbb{P}(A_j)$. Choose ϵ so that the resulting upper bound is at most δ .

- (10 pts) Let $\hat{j} \in \arg \min_j \hat{L}_j$ and $j^* \in \arg \min_j L_j$. Show that, on the simultaneous event from part 1,

$$L_{\hat{j}} \leq L_{j^*} + 2\epsilon.$$

Hint: compare $L_{\hat{j}}$ with $\hat{L}_{\hat{j}}$, then with $\hat{L}_{j^*}$, and finally with L_{j^*} .

- (5 pts) For a target excess risk $\eta > 0$, find a sufficient validation sample size to guarantee $L_{\hat{j}} - L_{j^*} \leq \eta$ with probability at least $1 - \delta$. How does this sufficient sample size change if M increases or η is halved?

2 Non-graded labs

2.1 Cross-validation

Exercise 1:

Implement 4-fold cross-validation.

- In this exercise, please fill in the notebook functions `cross_validation()` and `cross_validation_demo()`, and perform 4-fold cross-validation for polynomial degree 7. Plot the train and test RMSE as a function of λ . The resulting figure should look like Figure 1.
- How will you use 4-fold cross-validation to select the best model among various degrees, say from 2 to 10? Write code to do it in `best_degree_selection()`.

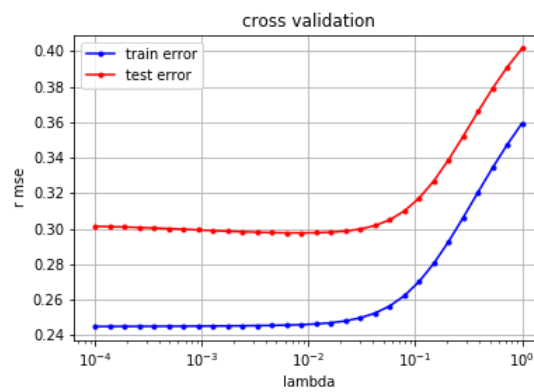


Figure 1: Effect of λ on training and test errors, calculated using 4-fold cross-validation

2.2 Visualizing the Bias-Variance Decomposition

Last lecture we introduced model selection, and we saw that the model complexity is crucial to the performance. In this problem, we will further investigate the effect of *model complexity* with the concept of *bias-variance decomposition*.

We will implement the figures seen in class representing the tradeoff and also seen in Figure 2 : for a big polynomial degree, the bias is small but the variance is large. The opposite is true for a small polynomial degree (however notice that the variance is still quite important). Choosing an intermediate degree leads to consistent predictions which are close to the true function we want to learn (optimal bias / variance tradeoff).

Exercise 2:

Visualizing the bias-variance trade-off.

- Complete the notebook function `bias_variance_one_seed()`: for 15 random datapoints, it finds the optimal fit (using the least square formula, with no regularisation λ) for a polynomial expansion of degree 1, 3 and 6.
- you can play around by changing the seed, the number of datapoints, the degree of the polynomial expansion etc.
- Now complete the notebook function `bias_variance_demo()` which performs many times the previous experiment but with a new random training set each time. You should obtain something similar to Figure 2.
- Comment the figures by explaining how the bias / variance tradeoff is shown in these plots.
- You can play around by changing the function you want to learn, the variance of the gaussian noise σ^2 , the degree of the polynomial expansion etc.

- **BONUS:** you can do similar figures but now you fix the degree of the polynomial expansion and add some regularisation λ . You will observe a similar bias / variance tradeoff when changing the magnitude of the regularisation.

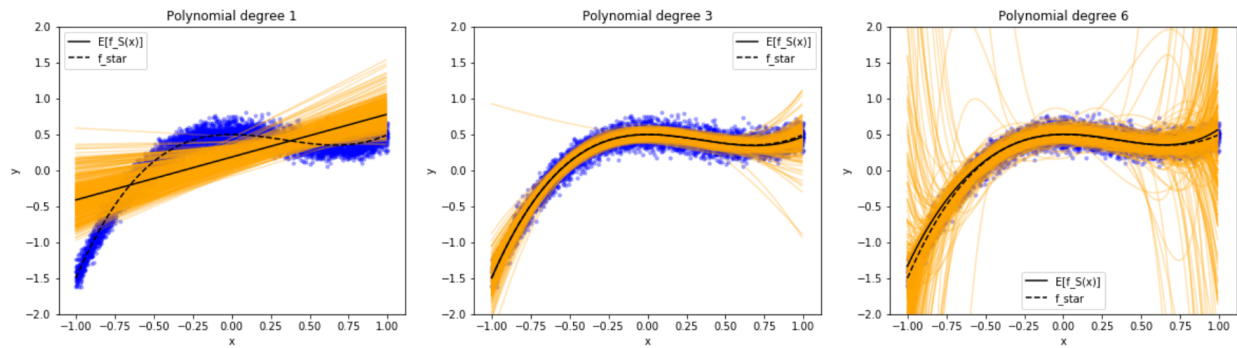


Figure 2: Visualizing the Bias-Variance Trade-off.